Kaustubh Patil . Jarvis Consulting

Data and Machine Learning Engineer with five years of experience building production AI/ML platforms across Azure, GCP, and AWS. At Gartner (Talent Neuron), designed Databricks pipelines on Azure processing 33TB of workforce data and serving 165M daily predictions with full MLOps governance, including drift detection, bias checks, and MLflow lineage. Skilled in Python, SQL, Kubernetes, Terraform, and GitOps CI/CD, with growing experience in GenAI platform work using LangChain, RAG pipelines, and LLM evaluation. Passionate about the engineering craft behind ML: reusable frameworks, observability, and reliable infrastructure. Currently expanding into cloud and data engineering through the Jarvis program and seeking a Machine Learning Engineer or MLOps Engineer role.

Skills

Proficient: Python (OOP), SQL, Databricks, MLflow, Apache Airflow, Docker, Kubernetes (GKE/AKS/EKS), Azure (ADF, AML, Synapse), GCP (Vertex AI, BigQuery, GKE), Terraform, GitHub Actions, ArgoCD, Git, Linux/Bash, Agile/Scrum

Competent: LangChain, LangGraph, RAG Pipelines, LLM Evaluation, LiteLLM, LangSmith, Apache Spark, Snowflake, Apache Kafka, Prometheus/Grafana, Splunk, REST APIs, GitOps, AWS (S3, Redshift, Lambda), PostgreSQL

Familiar: Scala, Java, JavaScript (React/Node.js), MongoDB, Cassandra, Elasticsearch, Datadog, Jenkins, Hadoop

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_Patil

Cluster Monitor [GitHub]: Designed and implemented a Linux Cluster Monitoring Agent in Bash to collect hardware specs and real-time resource usage from CentOS servers. Provisioned a PostgreSQL database in Docker for persistent storage of node metrics, and automated scheduled collection through crontab. Wrote SQL analytical queries to surface cluster utilization trends and capacity-planning insights, with the full agent and database containerized for reproducible deployment.

SQL Data Analytics [GitHub]: Designed a normalized PostgreSQL schema (1NF, 2NF, 3NF) to model a transactional dataset, and implemented data analytics through complex SQL queries involving joins, aggregations, subqueries, and window functions. Provisioned the database with Docker and explored it through DBeaver, producing reproducible analytical workflows and validating query results against expected business metrics.

Python Data Analytics [GitHub]: Developed an end-to-end analytical pipeline in Python on the U.S. CFPB Consumer Complaint Database (1.28M rows, 5,275 firms, 2011-2019). Performed data profiling, cleaning, and feature engineering with Pandas, including time-series, lag, and company-tier features. Built interactive Plotly dashboards covering market concentration, product mix shift, and a seven-dimension Bureau Accountability Scorecard. Validated outputs against published CFPB statistics and documented a Spark/Databricks migration roadmap.

Highlighted Projects

CI/CD Automation with DevOps Tools [GitHub]: Built a fully automated end-to-end CI/CD pipeline integrating Jenkins for orchestration, SonarQube for static code analysis, JFrog Artifactory for artifact management, and Docker with Kubernetes for containerized deployment. Instrumented the pipeline with Prometheus and Grafana for build-health monitoring and alerting, demonstrating a production-grade DevOps workflow from commit to deployment.

Scalable VPC Architecture on AWS [GitHub]: Designed and deployed a modular, scalable, and secure virtual network infrastructure on AWS using Amazon VPC. Implemented public and private subnets across multiple availability zones, configured NAT gateways, route tables, and security groups, and applied infrastructure-as-code principles to enable repeatable provisioning of a production-ready cloud network foundation.

Biological Network Analysis & Protein Kinase Inhibition in Breast Cancer [GitHub]: Led a peer-reviewed bioinformatics study on protein kinase inhibition networks in breast cancer. Designed the experimental approach, performed network analysis on protein interaction data, validated findings against published literature, and authored the manuscript as lead author through to publication.

Professional Experiences

Machine Learning Engineer, Jarvis (Jan 2026 present): Completing intensive training across Linux, SQL, core Java, Python, Spark, and cloud DevOps. Building production-grade projects using Git, Docker, and CI/CD pipelines while practicing agile workflows and peer code reviews. Strengthening foundational software engineering skills to complement existing ML and MLOps expertise, with a focus on transitioning into a Machine Learning Engineer or MLOps Engineer role.

Data & Research Associate (AI/ML Platform), Gartner (Talent Neuron) (Sep 2023 Jan 2026): Designed Databricks AI/ML pipelines on Azure processing 33TB of workforce data and serving 165M daily predictions. Developed ML orchestration with Azure Data Factory and Airflow, automating retraining and ensuring governance through drift detection, bias checks, and MLflow lineage. Deployed containerized ML services with Docker and Kubernetes, wrote Terraform IaC, and established GitOps CI/CD workflows using ArgoCD and GitHub Actions. Built reusable Python OOP frameworks adopted engineering-wide, and integrated Splunk and Prometheus/Grafana for monitoring, SLOs, and incident response. Collaborated with infrastructure and security teams to embed IAM, RBAC, and policy controls across multi-tenant AI/ML workflows.

Data Solutions Engineer, IPLIX Media (Jan 2022 Sep 2023): Built real-time analytics pipelines on Azure Data Lake and GCP (BigQuery, Cloud Functions) processing 500K weekly unstructured social media signals, accelerating decision-making by 35%. Orchestrated Docker and AKS environments through Terraform IaC and developed CI/CD pipelines for SQL, Python, and ML workflows using GitHub Actions, reducing deployment cycles by 40%. Designed scalable multi-source data schemas enabling automated creator performance scoring and AI-driven segmentation, with RAG-adjacent enrichment patterns, and automated data quality validation that reduced manual intervention by 25%.

Data Engineer, Mosil Lubricant (Aug 2020 Jan 2022): Built an Azure Synapse Analytics reporting platform that replaced manual validation workflows and reduced operational reporting effort by 30%. Developed Azure Data Factory pipelines for market intelligence extraction, improving signal identification accuracy by 15%. Implemented Azure SQL monitoring and alerting that ensured 99.9% data availability for executive dashboards, and partnered with business stakeholders to translate reporting requirements into scalable data models.

Education

York University (Sep 2025), Diploma, Cloud Operations

Conestoga College of Technology (Sep 2024), Diploma, Big Data Solutions Architecture

Dr. D. Y. Patil University (July 2023), Bachelor of Technology, Data Science and Bioinformatics - Lead author, peer-reviewed publication: Biological Network Analysis & Protein Kinase Inhibition in Breast Cancer

Miscellaneous

- DP-700 Microsoft Certified: Fabric Data Engineer Associate
- SnowPro Core Certified (In Progress)
- Open-source contributor in MLOps and GenAI tooling
- Continuous learning across Azure, GCP, and Snowflake certifications